

Chemistry Suggestion for Mid Examination by Siraj Sir (Updated)

Segment-01 Periodic Classification of Elements

1. State Mendeleev's and Modern Periodic Laws. Mention some properties of Periodic Table and classify the elements on the basis of electronic configuration.

Answer:

Mendeleev's Periodic Law states that *"The properties of elements are a periodic function of their atomic masses."* Mendeleev arranged the known elements in order of increasing atomic mass and grouped them according to similar chemical properties. He left blank spaces for elements yet to be discovered and accurately predicted their properties. However, some elements did not fit perfectly when arranged by mass, causing inconsistencies.

Modern Periodic Law, given by Henry Moseley, corrected this. It states that *"The properties of elements are a periodic function of their atomic numbers."* This law led to the modern periodic table where elements are arranged by increasing atomic number, which solved the placement anomalies of Mendeleev's table.

Some important **properties of the periodic table** include:

- It consists of **7 periods** (horizontal rows) and **18 groups** (vertical columns).
- Elements are arranged in **increasing order of atomic number**.
- It is divided into **blocks** based on the type of orbital the valence electron enters: **s-block**, **p-block**, **d-block**, and **f-block**.

Based on **electronic configuration**, elements are classified as:

- **s-block elements**: Groups 1 and 2, where the last electron enters the s-orbital.
- **p-block elements**: Groups 13 to 18, where the last electron enters the p-orbital.
- **d-block elements**: Transition metals (Groups 3 to 12), with electrons entering d-orbitals.
- **f-block elements**: Lanthanides and actinides, where electrons fill the f-orbitals.

2. "Electronic Configuration of the elements is the basis of the Periodic Table" – Explain.

Answer:

Electronic configuration refers to the arrangement of electrons in the orbitals of an atom. In the modern periodic table, elements are arranged in increasing order of atomic number, which also reflects the pattern of their electron configurations. The properties of elements repeat periodically because the pattern of electron arrangement repeats after a certain interval.

Elements in the same group have similar outer electron configurations, which gives them similar chemical properties. For example, all Group 1 elements (like Na, K, Rb) have one electron in their outermost s-orbital (ns^1), making them highly reactive metals. This shows that electronic configuration determines the position of an element in the periodic table and influences periodic trends such as ionization energy, atomic size, and electronegativity.

Therefore, the periodicity in properties of elements is due to the recurring patterns in their electronic configurations, making it the fundamental basis for the modern periodic table.

3. Determine the position of different elements in the periodic table.

Answer:

To determine the position of an element in the periodic table, we need to look at its **electronic configuration**:

- **Period number:** Given by the highest principal quantum number (n) in the electron configuration.
- **Group number:**
 - For **s-block elements**: Group number = number of valence electrons (1 or 2).
 - For **p-block elements**: Group number = 10 + number of valence electrons (i.e., 13 to 18).
 - For **d-block elements**: Group number is usually the sum of $(n-1)d$ and ns electrons.

Examples:

- **Sodium (Na):** $1s^2 2s^2 2p^6 3s^1 \rightarrow$ Outer shell is $3s^1 \rightarrow$ Period = 3, Group = 1 (s-block)
- **Carbon (C):** $1s^2 2s^2 2p^2 \rightarrow$ Outer shell is $2p^2 \rightarrow$ Period = 2, Group = 14 (p-block)

So, the electronic configuration tells us exactly where to place the element in the periodic table.

4. Why ionization energy of N is higher than O?

Answer:

Ionization energy is the energy required to remove an electron from a gaseous atom. Normally, we expect ionization energy to increase across a period. However, **nitrogen (N)** has a **higher ionization energy than oxygen (O)** due to its **stable half-filled p-orbital configuration** ($2p^3$).

Nitrogen's electronic configuration is $1s^2 2s^2 2p^3$, with three electrons in separate p-orbitals (each with parallel spins), giving it extra stability due to symmetrical distribution and exchange energy. In contrast, oxygen has a $2p^4$ configuration, with two electrons paired in one of the p-orbitals, which introduces repulsion and reduces stability.

As a result, it takes more energy to remove an electron from the more stable nitrogen atom than from oxygen, making nitrogen's ionization energy higher.

5. Why Fluorine (F) has the highest electronegativity?

Answer:

Fluorine (F) is the **most electronegative element** because of the following reasons:

- It has a **small atomic radius**, so the nucleus exerts a strong attractive force on bonding electrons.
- It has a **high effective nuclear charge**, meaning a strong pull from the nucleus on valence electrons.
- Fluorine is in **Group 17 (halogens)** and needs only **one electron to complete its octet**, making it highly eager to attract electrons.
- It has **low shielding effect**, which allows the nucleus to attract bonding electrons more effectively.

Due to all these factors, fluorine strongly attracts electrons in a chemical bond, giving it the **highest electronegativity value (3.98 on Pauling scale)**.

6. What is quantum number? Write four quantum numbers with their significance.

Answer:

Quantum numbers are values that describe the **energy, shape, orientation, and spin** of electrons in atoms. Each electron in an atom is described by a unique set of four quantum numbers:

1. **Principal Quantum Number (n)**: Indicates the main energy level or shell. ($n = 1, 2, 3, \dots$)
→ Higher n = higher energy level and distance from nucleus.
2. **Azimuthal Quantum Number (l)**: Defines the subshell or shape of orbital. ($l = 0$ to $n-1$)
→ $l = 0$ (s), 1 (p), 2 (d), 3 (f)
3. **Magnetic Quantum Number (m or m_l)**: Specifies the orientation of the orbital in space.
→ For $l = 1$ (p-orbital), $m = -1, 0, +1$
4. **Spin Quantum Number (s or m_s)**: Represents the direction of electron spin.
→ $s = +\frac{1}{2}$ or $-\frac{1}{2}$

Together, these quantum numbers uniquely describe the position and behavior of an electron within an atom.

7. Comparison Rutherford's Model with Bohr's Model. Which model is more acceptable and why?

Answer:

Rutherford's Atomic Model proposed that an atom consists of a dense, positively charged nucleus with electrons revolving around it, similar to planets orbiting the sun. This explained the existence of a nucleus but **couldn't explain** why electrons don't spiral into the nucleus due to attraction, nor the **atomic spectra**.

Bohr's Atomic Model improved upon this by introducing the idea that electrons revolve in **specific, stable orbits (energy levels)** without radiating energy. Electrons can jump between levels by absorbing or releasing energy, explaining the **line spectra of hydrogen**.

Bohr's model is more acceptable because:

- It explains **atomic stability**.
- It successfully explains the **line spectra** of hydrogen.
- It introduces **quantized energy levels**.

Although Bohr's model couldn't explain multi-electron atoms or fine spectral lines, it was a major advancement over Rutherford's model and laid the foundation for modern quantum theory.

Segment-02

Electronic Theory of Elements

1. What is Chemical Bond? Write types of chemical bond.

Answer:

A **chemical bond** is a force of attraction that holds atoms or ions together in a molecule or compound. Atoms bond to achieve a **stable electronic configuration**, usually an octet (8 electrons) in their outermost shell, similar to noble gases.

There are **three main types of chemical bonds**:

1. **Ionic Bond** – Formed by the transfer of electrons from a metal to a non-metal. One atom loses electrons and becomes a positively charged ion (cation), and the other gains electrons to become a negatively charged ion (anion). These oppositely charged ions attract each other.
2. **Covalent Bond** – Formed when two non-metal atoms **share electrons** to complete their octet.
3. **Coordinate (or Dative) Covalent Bond** – A type of covalent bond where both shared electrons come from the same atom.

There are also other types like **metallic bonds** (between metal atoms) and **hydrogen bonds** (intermolecular force between molecules), but the three above are most fundamental.

2. What do you mean by Covalent Bond? How covalent bond is formed? Explain briefly with a suitable example.

Answer:

A **covalent bond** is a chemical bond formed when **two atoms share one or more pairs of electrons**. It usually occurs between **non-metal atoms**, which cannot lose electrons easily.

Covalent bonds are formed to help atoms achieve **noble gas configuration (octet rule)**. Each shared pair of electrons constitutes one covalent bond.

Example: In a water molecule (H_2O), each hydrogen atom shares one electron with oxygen, and oxygen shares one of its electrons with each hydrogen, forming two single covalent bonds:



Here, the shared pairs make the oxygen atom stable with 8 electrons in its outer shell and each hydrogen with 2 electrons (duplet), thus forming a **stable covalent molecule**.

3. How ionic bond is formed? Explain briefly how bond formation occurred in NaCl.

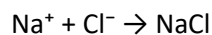
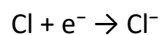
Answer:

An **ionic bond** is formed when **one atom donates electrons** and another **accepts those electrons**, resulting in positively and negatively charged ions that attract each other due to **electrostatic force**.

In sodium chloride (NaCl):

- Sodium (Na), a metal, has the electronic configuration: 2,8,1. It loses 1 electron to form Na^+ ion.
- Chlorine (Cl), a non-metal, has the configuration: 2,8,7. It gains 1 electron to complete its octet and becomes Cl^- .

This electron transfer creates **Na^+ and Cl^- ions**, and the strong attraction between them forms an **ionic bond**:



The result is a stable ionic compound with a regular crystalline structure.

4. What do you mean by Coordination Covalent Bond? Explain with an example.

Answer:

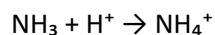
A **coordination covalent bond** (also called a **dative bond**) is a type of covalent bond where **both electrons in the shared pair come from the same atom**.

This usually happens when one atom has a **lone pair of electrons** and the other atom has an **empty orbital** to accept it.

Example: In the formation of **ammonium ion (NH_4^+)**:

- Ammonia (NH_3) has a lone pair of electrons on nitrogen.
- A hydrogen ion (H^+) has no electrons but needs 2 to become stable.
- Nitrogen donates its lone pair to H^+ , forming a coordinate bond.

Representation:



The new N–H bond is a **coordinate bond**, though it behaves like any regular covalent bond after formation.

5. What is Hybridization? Explain Hybridization process in BeCl_2 / BCl_3 / H_2O .

Answer:

Hybridization is the process of **mixing atomic orbitals** of an atom to form new **hybrid orbitals** that are equivalent in energy and suitable for bonding. It explains the geometry and bond angles in molecules.

(i) BeCl_2 (Beryllium Chloride)

- Beryllium's configuration: $1s^2 2s^2 \rightarrow$ needs to form 2 bonds.
- It undergoes **sp hybridization**: one 2s and one 2p orbital mix to form 2 sp hybrid orbitals.
- Geometry: **Linear**, Bond angle: **180°** .

(ii) BCl_3 (Boron Trichloride)

- Boron's configuration: $1s^2 2s^2 2p^1 \rightarrow$ forms 3 bonds.
- It undergoes **sp^2 hybridization**: one 2s and two 2p orbitals mix to form 3 sp^2 hybrid orbitals.
- Geometry: **Trigonal planar**, Bond angle: **120°** .

(iii) H_2O (Water)

- Oxygen's configuration: $1s^2 2s^2 2p^4 \rightarrow$ has 2 lone pairs and forms 2 bonds.
- It undergoes **sp^3 hybridization**: one 2s and three 2p orbitals mix to form 4 sp^3 hybrid orbitals.
- Two are bonding pairs (with H), and two are lone pairs.
- Geometry: **Bent/V-shape**, Bond angle: **104.5°** due to lone pair repulsion.

Segment-03

Chemistry of Transition Elements, Lanthanides & Actinides

1. What is Transition Elements? Explain why all d-block elements aren't transition elements but all transition elements are d-block elements?

Answer:

Transition elements are the elements found in the **d-block of the periodic table**, and they typically have **partially filled d-orbitals** either in their elemental state or in one of their common oxidation states.

The general definition:

“Transition elements are those d-block elements that have an incomplete d-subshell in their atoms or in their most common oxidation states.”

Why all d-block elements aren't transition elements:

Some d-block elements, like **Zinc (Zn)**, **Cadmium (Cd)**, and **Mercury (Hg)**, have **completely filled d-orbitals** (e.g., $\text{Zn} = 3d^{10} 4s^2$), both in elemental and ionic states ($\text{Zn}^{2+} = 3d^{10}$). Since they do not have **incomplete d-subshells**, they are **not considered true transition elements**.

But all transition elements are d-block elements because they all belong to the d-block and meet the required condition of having incomplete d-orbitals.

2. What are the properties of transition elements? Explain briefly three of them.

Answer:

Transition elements show unique properties due to their electronic structure. Some key properties include:

1. Variable Oxidation States:

Transition metals can lose different numbers of electrons due to the close energy levels of (n-1)d and ns orbitals.

Example: Iron shows +2 (Fe^{2+}) and +3 (Fe^{3+}) oxidation states.

2. Formation of Colored Compounds:

Many transition metal compounds are colored due to **d-d electron transitions** when they absorb specific wavelengths of light.

Example: Cu^{2+} (blue), Fe^{3+} (yellow-brown), Ni^{2+} (green).

3. Catalytic Properties:

Transition metals and their compounds often act as **catalysts** in chemical reactions due to their

variable oxidation states and surface properties.

Example: Iron in Haber process, Vanadium(V) oxide in Contact process.

Other properties include high melting/boiling points, magnetic behavior, and the formation of complex ions.

3. What do you mean by Lanthanides & Actinides series? Write some properties of them.

Answer:

Lanthanides and Actinides are two series of **f-block elements** placed separately at the bottom of the periodic table to maintain its structure and avoid excessive widening. These elements are characterized by the filling of electrons in the f-orbitals.

- **Lanthanides (Ln):** Elements with atomic numbers **57 to 71** (from **Lanthanum, La** to **Lutetium, Lu**). These are known as the **4f series**, as their differentiating electron enters the 4f orbital. Though Lanthanum does not have a 4f electron, it is included in the series due to its similar properties.
- **Actinides (An):** Elements with atomic numbers **89 to 103** (from **Actinium, Ac** to **Lawrencium, Lr**). This group is called the **5f series** since their differentiating electron enters the 5f orbital. Like La, Actinium is also included based on chemical similarities despite having no 5f electron.

Properties of Lanthanides:

- All are **metals**, typically **soft, silvery-white**, and have **high melting points**.
- They commonly show a **+3 oxidation state**, though +2 and +4 may occur in some cases.
- They exhibit **lanthanide contraction**: a gradual decrease in atomic and ionic radii across the series.
- Most lanthanide ions are **paramagnetic** due to the presence of **unpaired f-electrons**.
- They often form **colored compounds**, useful in analytical and industrial applications.
- They are used in **strong permanent magnets, phosphors in TV screens and LEDs, lasers**, and **nuclear applications**.

Properties of Actinides:

- **All actinides are radioactive**, with heavier elements being mostly **synthetic** and short-lived.
- They display a wide range of **oxidation states**, commonly from **+3 to +6**, unlike lanthanides.
- They have **large atomic and ionic sizes**, which **decrease across the series** (actinide contraction).
- Actinides are **electropositive** and **chemically reactive**, especially with halogens and water.
- They form **complex compounds** and exhibit **variable magnetic and spectral behaviors**.
- Some actinides like **Uranium (U)** and **Plutonium (Pu)** are crucial in **nuclear reactors and weapons**.

4. What do you mean by Lanthanides & Actinides contraction?

Answer:

Lanthanide contraction refers to the **gradual decrease in atomic and ionic radii** of the lanthanide elements from **Lanthanum (La) to Lutetium (Lu)** as the atomic number increases. This occurs despite the addition of electrons to the same 4f orbital.

- The main reason behind this contraction is the **poor shielding effect of 4f electrons**. As the nuclear charge increases with each added proton, the ineffective shielding by 4f electrons allows the nucleus to pull the electron cloud closer, reducing the size of the atom/ion.

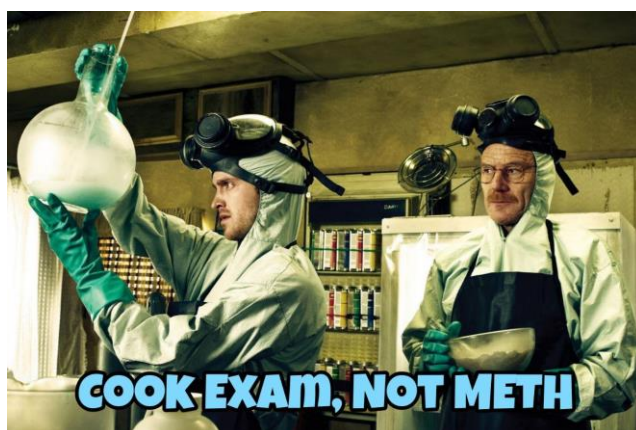
Effects of lanthanide contraction:

- It leads to **similarity in atomic sizes** of elements in the 4d and 5d series (e.g., **Zr and Hf**).
- It **complicates the chemical separation** of lanthanides due to their similar ionic radii.
- It affects the **basicity, solubility, and complex formation tendencies** of their compounds.

Actinide contraction is a **similar decrease in atomic and ionic sizes** observed across the actinide series from **Actinium (Ac) to Lawrencium (Lr)**. However, this contraction is **less regular and more pronounced** due to additional factors such as:

- The stronger relativistic effects acting on 5f electrons.
- The presence of **radioactive instability** and **variable oxidation states**, which influence electron distribution and effective nuclear charge.

Both lanthanide and actinide contractions have significant impacts on the **chemistry, bonding, and applications** of f-block elements.



Disclaimer: This is a basic guide, not a guaranteed exam prep source. Please use it alongside past question papers. Review the information carefully as some parts may be inaccurate.